SwinSUNet: Pure Transformer Network for Remote Sensing Image Change Detection

Cui Zhang, Liejun Wang, Shuli Cheng, and Yongming Li

***Abstract*— Convolutional neural network (CNN) can extract effective semantic features, so it was widely used for remote sensing image change detection (CD) in the latest years. CNN has acquired great achievements in the field of CD, but due to the intrinsic locality of convolution operation, it could not capture global information in space-time. The transformer was proposed in recent years and it can effectively extract global information, so it was used to solve computer vision (CV) tasks and achieved amazing success. In this article, we design a pure transformer network with Siamese U-shaped structure to solve CD problems and name it SwinSUNet. SwinSUNet contains encoder, fusion, and decoder, and all of them use Swin transformer blocks as basic units. Encoder has a Siamese structure based on hierarchical Swin transformer, so encoder can process bitemporal images in parallel and extract their multiscale features. Fusion is mainly responsible for the merge operation of the bitemporal features generated by the encoder. Like encoder, the decoder is also based on hierarchical Swin transformer. Different from the encoder, the decoder uses upsampling and merging (UM) block and Swin transformer blocks to recover the details of the change information. The encoder uses patch merging and Swin transformer blocks to generate effective semantic features. After the sequential process of these three modules, SwinSUNet will output the change maps. We did expensive experiments on four CD datasets, and in these experiments, SwinSUNet achieved better results than other related methods.**

***Index Terms*— Change detection (CD), deep learning, remote sensing image, transformer.**

I. Introduction

REMOTE sensing image change detection (CD) is a technology that generates change maps by comparing and analyzing the multitemporal or bitemporal images. In the binary change map, the value of every pixel is 1 or 0, which represents whether the corresponding position has changed or not. CD technology is widely used in urban expansion [1], [2], ecosystem monitoring [3], land use CD [4], damage assessment [5], environmental monitoring [6], and so on. Automatic CD technology greatly promoted the development of these applications, and thus, it has attracted the attention of many scholars in recent years.

During the latest years, convolutional neural network (CNN) has acquired great achievements in many subfields of computer vision (CV), such as image classification [7]–[9], object detection [10]–[12], semantic segmentation [13], [14], and face recognition [15]–[17]. In the meantime, many excellent CNN-based CD algorithms have been proposed [18]–[27]. Zhan *et al.* [18] introduced the Siamese convolutional network into the CD task for the first time, which can process bitemporal images in parallel. Since then, there have been many CD methods based on the Siamese structure. Daudt *et al.* [20] designed three effective CD architectures based on fully convolutional neural networks (FCNNs), fully convolutional early fusion (FC-EF), fully convolutional Siamese-concatenation (FC-Siam-conc), and fully convolutional Siamese-difference (FC-Siam-diff). However, due to the intrinsic locality of convolution operation, it is still difficult for these methods to extract the global information in space-time. Some methods have been proposed to solve this problem. Chen and Shi [21] and Chen *et al.* [22] used ResNet [8] as the backbone network, Zhang *et al.* [23] used the dilated convolution to replace the traditional convolution, and Chen and Shi [21], Chen *et al.* [22], Fang *et al.* [25], and Peng *et al.* [26] used attention mechanism, and these methods increase the reception field of the network to some extent, but the ability of extracting global information is still limited.

The transformer [28] was proposed in 2017, originally used for natural language processing; recently, many scholars have introduced transformer into the field of CV [29]–[40]. Due to the amazing semantic representation capability of the transformer, it performs as well or even better than CNN in various CV tasks, such as image classification [29]–[32], object detection [33]–[35], semantic segmentation [31], [33], [36], image generation [37], [38], and super-resolution [39], [40]. Unlike previous approaches that either use the attention module and CNN together or use the attention module to replace a specific component in CNN, the vision transformer (ViT) [29] is the first pure transformer approach for image classification tasks. Taking 2-D image patches with positional embeddings as inputs and pretraining on large dataset, ViT achieved comparable performance with the CNN-based methods. Liu *et al.* [32] proposed a hierarchical Swin transformer, which used shifted windowing scheme to calculate self-attention in a local window. The shifted windowing scheme reduced the computational complexity of self-attention. To its credit,

Manuscript received July 15, 2021; revised January 29, 2022 and March 5, 2022; accepted March 13, 2022. Date of publication March 16, 2022; date of current version April 12, 2022. This work was supported by the Tianshan Innovation Team of Xinjiang Uygur Autonomous Region under Grant 2020D14044 and in part by the National Science Foundation of China under Grant U1903213. *(Corresponding author: Liejun Wang.)*

Cui Zhang, Liejun Wang, and Yongming Li are with the College of Information Science and Engineering, Xinjiang University, Ürümqi 830046, China (e-mail: zhangcui2020@stu.xju.edu.cn; wljxju@xju.edu.cn; lym@xju.edu.cn).

Shuli Cheng is with the College of Information Science and Engineering and the College of Mathematics and System Science, Xinjiang University, Ürümqi 830046, China (e-mail: cslxju@xju.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3160007

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

Swin transformer not only improved speed but also improved accuracy, which achieved the optimal performance in several CV tasks, such as image classification, object detection, and semantic segmentation. Due to the success of ViT and Swin transformer in image classification, we can anticipate the development potential of the transformer in the field of CD.

Inspired by Swin transformer's success in image classification, we design a new network named SwinSUNet, hoping to solve the CD task by leveraging Swin transformer. The bitemporal CD task is different from other tasks, its input is usually a pair of bitemporal images, and its output is dense prediction. Therefore, we adopt a pure Swin transformer network with Siamese U-shaped structure to solve the CD task. The basic unit of the previous CNN-based CD network is convolution operation, while the basic unit of SwinSUNet is Swin transformer block, so SwinSUNet is not limited by the intrinsic locality of convolution operation and can capture global information in space-time.

SwinSUNet includes three parts: encoder, fusion, and decoder. In order to process bitemporal images, we design an encoder with a Siamese structure. Encoder first transforms the input image into image tokens by using patch partition and linear embedding. Subsequently, the encoder generates multiscale features by using a hierarchical Swin transformer, which is composed of patch merging and Swin transformer blocks. For the bitemporal features generated by encoder, fusion connects them by using concatenation, linear projection, and Swin transformer blocks. The decoder first uses a hierarchical Swin transformer to progressively predict change information. Unlike encoder, hierarchical Swin transformer in decoder consists of upsampling and merging (UM) block and Swin transformer blocks. With the processing of each stage of hierarchical Swin transformer, the resolution of feature map is gradually expanded, and the change information is gradually generated. The decoder finally uses an upsampling layer and a linear projection layer to restore the change map with the same resolution as the input images.

To the authors' knowledge, SwinSUNet is the first pure transformer network for CD task. Compared with the previous CNN-based methods, SwinSUNet can better extract the global information in space-time.

The rest of this article is organized as follows: In Section II, we describe the related CD algorithms. In Section III, we introduce the overall framework of SwinSUNet and describe each part of SwinSUNet in detail. In Section IV, we compare SwinSUNet with other state-of-the-art algorithms through experiments and confirm the effectiveness of SwinSUNet. In Section IV, we summarize SwinSUNet's performance.

II. Related Work

A. CNN-Based CD Methods

Because of the strong feature representation capability of CNN, many previous CD methods are based on CNN. CNN-based CD methods usually enhance the semantic representation ability of network by changing the network structure, optimizing loss function, adding attention mechanism, and so on.

In terms of network structure, Zhan *et al.* [18] used a Siamese convolutional network to solve the CD task for the first time. With the Siamese structure, we can simultaneously process the bitemporal images, and most current CD methods are based on this structure. Daudt *et al.* [20] designed the first end-to-end training CD method and proposed three effective FCNN-based architectures. In [20], FC-EF concatenates the bitemporal images as the input of the network, while FC-Siam-conc and FC-Siam-diff leverage a Siamese structure, which can directly process bitemporal images. The most intuitive way to reduce the inherent locality of convolution operation is to increase the reception field. Therefore, Zhang *et al.* [23] used the dilated convolution instead of the traditional convolution and achieved certain results. Chen and Shi [21] proposed STANet and Chen *et al.* [22] proposed DASNet, and they used Resnet18 and Resnet50 as the backbone networks. Compared with the shallow networks, STANet and DASNet have stronger feature extraction capability. Everyone knows that dense connections [9] between features can improve the network performance. Fang *et al.* [25] and Peng *et al.* [26] added dense connections between the features of different layers, so as to enhance the capability of the CD network.

In the remote sensing images, the changed pixels are far less than the unchanged pixels, so there exists a serious data imbalance in the CD task. Many scholars solved this problem by optimizing the loss function so that the changed and unchanged pixels participate in the loss calculation in the same proportion. Zhan *et al.* [18] proposed a weighted contrastive loss function, which increased the weight of the changed pixels in the loss calculation. STANet [21] and DASNet [22] further optimized the weighted contrastive loss function, and they proposed batch-balanced contrastive loss function and weighted double-margin contrastive loss function. Triplet loss [41] has a strong classification capability, which focuses on both intraclass compaction and interclass separability. Zhang *et al.* [23] introduced triplet loss into the CD task. In the calculation process of triplet loss, the changed and unchanged pixels appear in balance, which solves the data imbalance and achieves better CD results. In SNUNet proposed by Fang *et al.* [25], a hybrid loss function was used to optimize the network.

Since attention mechanism has achieved remarkable results in the CV task, many scholars have introduced attention mechanism into CD task. Liu *et al.* [24] and Chen *et al.* [22] introduced a dual attention module into the CD task to learn features that contain both channel information and spatial information. Fang *et al.* [25] proposed an ensemble channel attention module (CAM) to fuse the features of various levels, so as to generate stronger change features. In order to detect changes with different sizes, Chen and Shi [21] proposed a pyramid spatial–temporal attention module, which can extract features from different scales. Peng *et al.* [26] designed a dense attention method to extract richer and more effective features. Chen *et al.* [27] introduced the transformer into the field of CD for the first time and proposed bitemporal image transformer (BiT). First of all, BiT uses CNN to generate semantic features. Second, BiT leverages a transformer module

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

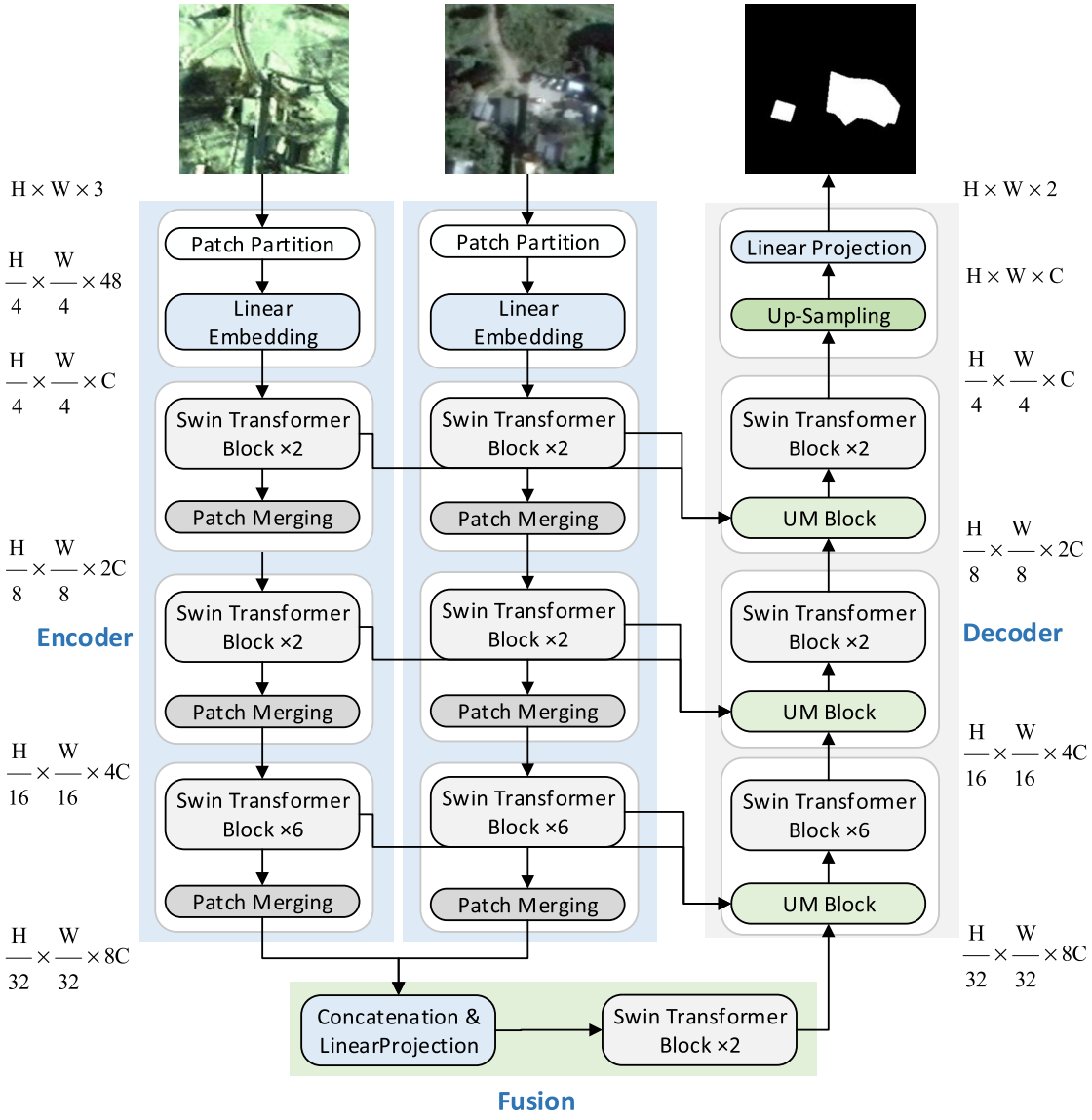


Fig. 1. Framework overview of SwinSUNet.

to further process CNN features. Finally, BiT uses a prediction head module to generate the change maps.

Although the above methods have improved the ability of CD network to a certain extent, due to the inherent locality of convolution operation, the CNN-based methods cannot effectively extract long-term global features, thus limiting the ability of CD network. Unlike the previous methods, this article attempts to explore the potential of pure transformer network for the CD task.

B. Transformer-Based Vision Backbones

ViT [29] was proposed in 2020, and it is the first pure transformer network for CV. In image classification, ViT achieves comparable results with CNN-based algorithms. However, one drawback of ViT is that we need to pretrain ViT on a larger dataset, which makes the training of ViT inconvenient. DeiT [30] proposed some training strategies to solve this drawback, and these strategies enable ViT to be trained directly on ImageNet without pretraining.

The computational complexity of ViT is quadratic to the size of the input image, so ViT is not suitable for the dense vision tasks. Swin transformer [32] used shifted windowing scheme to calculate self-attention in a local window, which not only reduced the computational complexity but also acquired the best results in several CV tasks. Inspired by the success of Swin transformer in dense vision tasks, we introduce the Swin transformer into the CD task and propose SwinSUNet that is a pure Swin transformer network with a Siamese U-shaped structure.

III. Methodology

A. Framework Overview

The framework of SwinSUNet is shown in Fig. 1. From this figure, we can see that SwinSUNet has a similar structure to FC-Siam-conc [20], and both of them include an encoder

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

module with a Siamese structure and a decoder module that is symmetric to the encoder. The difference between these two algorithms is that the basic unit of FC-Siam-conc is convolution operation, while the basic unit of SwinSUNet is Swin transformer block [32], so SwinSUNet can better extract global information in space-time.

SwinSUNet contains encoder, fusion, and decoder. The inputs of SwinSUNet are bitemporal images, and the outputs are change maps. Encoder has a Siamese structure, which can process bitemporal images in parallel and output bitemporal features. The encoder first uses patch partition and linear embedding to convert the bitemporal images into image tokens and then uses a hierarchical Swin transformer to generate multiscale features. In the fusion module, we mainly use concatenation, linear projection, and Swin transformer blocks to fuse the bitemporal features generated by the encoder. The decoder mainly leverages the features generated by the encoder and fusion to generate the change maps. First, the decoder uses a hierarchical Swin transformer to gradually restore the change information. Second, the decoder uses upsampling to expand the feature map to the same size as the original input images. In the end, the decoder leverages linear projection to reduce the feature dimension to 2, and at this time, the change maps are generated.

B. Swin Transformer

To reduce the computational complexity of global self-attention, Swin transformer [32] designed a Swin transformer block. In this block, standard multihead self-attention (MSA) module is replaced by (shifted) window-based multihead self-attention [(S)W-MSA] module. Besides (S)W-MSA module, the Swin transformer block includes multilayer perceptron (MLP) module, LayerNorm (LN) layers, and residual connections.

1) W-MSA Module: ViT [29] and DeiT [30] adopt the standard MSA module, which performs global self-attention calculation on the whole feature map. Their computational complexity is in a square relationship with the input size, which has a large computational overhead. In order to reduce the computational overhead, Swin transformer [32] proposed the W-MSA module. The W-MSA module only performs local self-attention calculation in each window, making the computational complexity linear to the input size.

The W-MSA module divides the feature map into several windows of size $M \times M$, as shown in Fig. 2(a). This module calculates self-attention separately in each window. The formula for calculating self-attention in the W-MSA module is the same as in the standard MSA module.

2) SW-MSA Module: Because the W-MSA module performs self-attention only in local windows, the receptive field of the network is limited. To enlarge the receptive field, Liu *et al.* [32] proposed the SW-MSA module.

The window partition strategy of SW-MSA module is shown in Fig. 2(b). The SW-MSA module window is offset by $(M/2), (M/2)$ relative to the W-MSA module window. In the Swin transformer, the W-MSA module alternates with the SW-MSA module to increase the network receptive field.

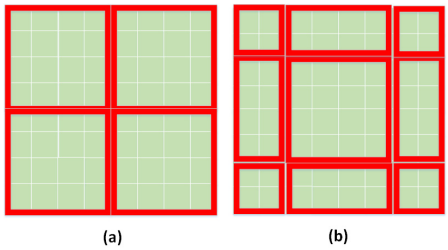


Fig. 2. (a) Window partition strategy of W-MSA module. (b) Window partition strategy of SW-MSA module.

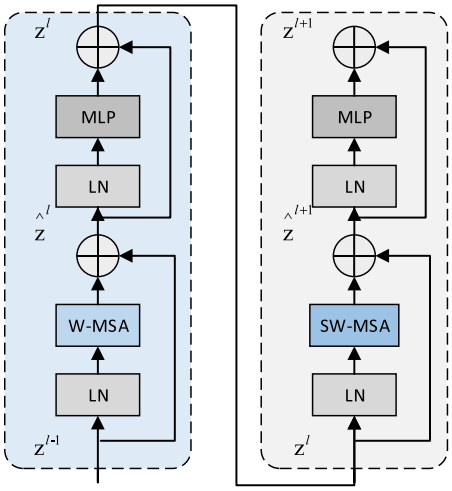


Fig. 3. Two successive Swin transformer blocks.

3) Self-Attention: Unlike the previous methods [29], the Swin transformer calculates self-attention in a local window but uses the same formula for calculating self-attention

$$
\begin{aligned}
Q &= TW^q \\
K &= TW^k \\
V &= TW^v \\
\text{Attention}(Q, K, V) &= \text{SoftMax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V \qquad (1)
\end{aligned}
$$

where $T \in \mathbb{R}^{M^2 \times d}$ represents the input of self-attention and $W^q, W^k, W^v \in \mathbb{R}^{d \times d}$ represent the weights of three linear projection layers. In this article, unless otherwise noted, the linear projection layer is implemented using a fully connected layer. $Q, K, V \in \mathbb{R}^{M^2 \times d}$ represent the query, key, and value matrices, respectively. M^2 denotes the resolution of $T, Q, K,$ and V and d denotes the channel dimension. In the Swin transformer, a bias matrix $\hat{B} \in \mathbb{R}^{(2M-1)\times(2M-1)}$ is parameterized, and the values of B are taken from this bias matrix.

4) Swin Transformer Block: Two successive Swin transformer blocks are shown in Fig. 3: the first one uses a W-MSA module and the other uses an SW-MSA module. As we described earlier, the W-MSA and SW-MSA modules use different window partition strategies to reduce the impact of window locality. The calculation formulas of two successive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

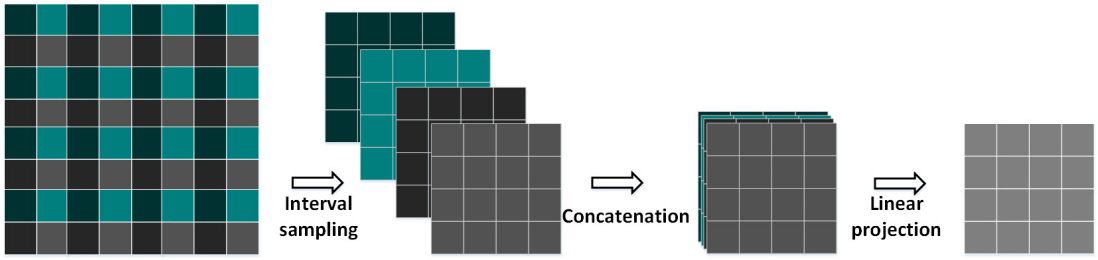


Fig. 4. Patch merging layer. For the sake of illustration, we have omitted the channel dimension in this diagram.

Swin transformer blocks are given as follows:

$$\begin{aligned}\hat{z}^l &= \mathrm{WMSA}\big(\mathrm{LN}\big(z^{l-1}\big)\big) + z^{l-1}\\ z^l &= \mathrm{MLP}\big(\mathrm{LN}\big(\hat{z}^l\big)\big) + \hat{z}^l\\ \hat{z}^{l+1} &= \mathrm{SWMSA}\big(LN\big(z^l\big)\big) + z^l\\ z^{l+1} &= \mathrm{MLP}\big(LN\big(\hat{z}^{l+1}\big)\big) + \hat{z}^{l+1}\end{aligned} \tag{2}$$

where $\hat{z}^l$ represents the output of (S)W-MSA module of the lth block and z^l represents the output of MLP module.

C. Encoder

Like ViT [29] and Swin transformer [32], SwinSUNet needs to transform the input image $X \in R^{H\times W\times C}$ into image tokens before extracting semantic features. In this process, SwinSUNet is the same as Swin transformer [32], including a patch partition module and a linear embedding module.

The patch partition module is responsible for converting the image into multiple tokens, and the linear embedding module is responsible for mapping the channel number of each token to a specified dimension. First, the patch partition module cuts the image into small patches. In our experiment, the resolution of each patch is 4×4 and the channel number is 3 (RGB). The total number of patches is $(H/4) \times (W/4)$. Each image patch can be considered a token. Then, the patch partition module flattens each token from 3-D data to 1-D data. The size of each token changes from $4 \times 4 \times 3$ to 48. Finally, the linear embedding module maps the channel number of each token from 48 to C. Now, the size of the feature map is $(H/4)\times(W/4)\times C$, and each spatial point of the feature map is a token. In our code, we used a convolution operation to realize both patch partition module and linear embedding module just like Swin transformer [32]. The input channel number of the convolution operation is 3, and the output channel number is C. The size of the convolution kernel is 4, and the stride is also 4.

Encoder further processes image tokens by using a hierarchical Swin transformer. The hierarchical Swin transformer consists of three stages, and each stage is composed of several Swin transformer blocks and one patch merging layer. The Swin transformer block is responsible for learning effective features. The patch merging layer is responsible for downsampling.

Like Swin transformer [32], we also use the patch merging layer (Fig. 4) to complete the downsampling operation. First, a patch merging layer performs interval sampling for the input feature $X_{E_i}^p \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times 2^{i-1}C}$ (i denotes the ith stage) and obtains four feature maps with a size of $(H/2^{i+2}) \times (W/2^{i+2}) \times 2^{i-1}C$. Then, we concatenate the four feature maps at the channel dimension to generate a feature map with a size of $(H/2^{i+2}) \times (W/2^{i+2}) \times 2^{i+1}C$. Finally, we use a linear projection layer to change the channel number from $2^{i+1}C$ to 2^iC. At this time, the size of the generated feature map $X_{E_i}^{p'}$ is $(H/2^{i+2}) \times (W/2^{i+2}) \times 2^iC$.

At each stage, the resolution of the feature map is halved, while the dimension is doubled. After three stages, the feature resolution is reduced from $(H/4) \times (W/4)$ to $(H/32) \times (W/32)$, and the dimension is increased from C to $8C$.

At each stage of encoder, the output feature maps of the Swin transformer blocks are saved and denoted as $X_{E_i}^1$ and $X_{E_i}^2$, where i stands for the ith stage. $X_{E_i}^1$ and $X_{E_i}^2$ will be input to the corresponding decoder stage. They will be used together with decoder features to recover the change information. In addition, the encoder finally outputs the bitemporal features X_E^1 and X_E^2, and these two feature maps will be input to the fusion module. It is easy to see that multiscale features can be generated by encoder, in which the high-level features hold strong semantic information and the low-level features retain the detailed information.

D. Fusion

Fusion is mainly used to merge the bitemporal features X_E^1 and X_E^2 generated by encoder, so as to facilitate the generation of the change map by the decoder. Fusion is composed of one concatenation layer, one linear projection layer, and two Swin transformer blocks. The concatenation layer is responsible for the connection of X_E^1 and X_E^2, the linear projection layer is responsible for the reduction of the feature dimension, and the Swin transformer blocks are responsible for the effective fusion of the features. During the process of the fusion module, the resolution of the feature map remains the same.

E. Decoder

It first uses a hierarchical Swin transformer to predict change information. Decoder's hierarchical Swin transformer also consists of three stages, and each stage includes one UM block and several Swin transformer blocks. The UM block is responsible for the upsampling and concatenation of the features. The Swin transformer block is responsible for generating effective change information.

We denoted the input feature map of each stage of decoder as X_{D_i}, where i stands for the ith stage. The channel dimension

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

of X_{D_i} is half of $X_{D_{i-1}}$ one, and its spatial resolution is twice the $X_{D_{i-1}}$ one. In these three stages, the change information has been gradually generated. After three stages, the feature resolution is increased from $(H/32) \times (W/32)$ to $(H/4) \times (W/4)$, and the dimension is decreased from $8C$ to C.

After the process of the hierarchical Swin transformer, the decoder first expands the feature map to the same resolution as the original input images by using an upsampling layer. Then decoder transforms the feature dimension from C to 2 by using a linear projection layer; at that time, the change map is predicted.

1) UM Block: It is composed of an upsampling block and a merging block, as shown in Fig. 5. First, UM block uses a patch reshaping block to realize the upsampling operation. Then, a merging block is used to fuse the upsampled feature map $X^U_{D_i}$ and the feature maps $X^1_{E_{3-i+1}}$ and $X^2_{E_{3-i+1}}$ generated by the encoder. Sections III-E2 and III-E3 will introduce patch reshaping block and merging block in detail, respectively.

2) Patch Reshaping: In the previous algorithms, the commonly used upsampling methods include bilinear interpolation and transposed convolution. In [42], a new upsampling method, patch expanding, was proposed. Patch expanding is composed of one linear projection layer and one rearrange layer, as shown in Fig. 6(a). The linear projection layer is used to increase the dimension of $X_{D_i} \in \mathbb{R}^{(H/2^{6-i}) \times (W/2^{6-i}) \times 2^{4-i}C}$ from $2^{4-i}C$ to $2^{5-i}C$. Then, the rearrange operation layer is used to enlarge the feature resolution from $(H/2^{6-i}) \times (W/2^{6-i})$ to $(H/2^{5-i}) \times (W/2^{5-i})$ and reduce the dimension from $2^{5-i}C$ to $2^{3-i}C$. The output of patch expanding is $X^U_{D_i} \in R^{(H/2^{5-i}) \times (W/2^{5-i}) \times 2^{3-i}C}$.

We make a small change to patch expanding and name the new upsampling method as patch reshaping. Patch reshaping also includes one linear projection layer and one rearrange layer, as shown in Fig. 6(b). In the linear projection layer, unlike patch expanding, the dimension of X_{D_i} not increases but remains the same. In the rearrange operation layer, the feature resolution is enlarged from $(H/2^{6-i}) \times (W/2^{6-i})$ to $(H/2^{5-i}) \times (W/2^{5-i})$ and the dimension is reduced from $2^{4-i}C$ to $2^{2-i}C$. The output of patch reshaping is $X^U_{D_i} \in \mathbb{R}^{(H/2^{5-i}) \times (W/2^{5-i}) \times 2^{2-i}C}$. We can find that in patch reshaping, the product of height, width, and dimension of feature map does not change.

The computational complexity of patch reshaping is less than patch expanding, and through experiments, we find that patch reshaping can achieve better results than patch expanding in the CD task. This is because the linear projection operation in patch reshaping does not increase the dimension of feature map, and the output of patch reshaping contains less redundant information than patch expanding, which is more conducive to the extraction of change information.

3) Merging Block: As the input feature of the decoder has gone through several downsampling operations, some spatial and detail information is lost. Some previous methods, such as FC-EF, FC-Siam-conc, and FC-Siam-diff [20], introduce $X^*_{E_{3-i+1}}$ that is generated by an encoder into decoder, thus mitigating information loss. We also adopt this technique and use a merging block to merge $X^1_{E_{3-i+1}}$, $X^2_{E_{3-i+1}}$, and $X^U_{D_i}$. However, different from previous methods, we add a CAM [43] to the merging block. A detailed implementation of the merging block is shown in Fig. 5.

First, the merging block uses a concatenation layer to fuse the upsampled feature map $X^U_{D_i}$ and the feature maps $X^1_{E_{3-i+1}}, X^2_{E_{3-i+1}} \in \mathbb{R}^{(H/2^{5-i}) \times (W/2^{5-i}) \times 2^{3-i}C}$ generated by the encoder. At this time, the concatenated feature map is denoted as $X^{\text{cat}}_{D_i}$, and its dimension is $(H/2^{5-i}) \times (W/2^{5-i}) \times (2^{2-i} \times 5)C$.

Second, a CAM is used to further process $X^{\text{cat}}_{D_i}$. As $X^{\text{cat}}_{D_i}$ is generated by the concatenation of $X^1_{E_{3-i+1}}$, $X^2_{E_{3-i+1}}$, and $X^U_{D_i}$ in the channel dimension, the CAM is added to better fuse them in the channel dimension.

Finally, the merging block uses the linear projection layer to reduce the feature map dimension from $(2^{2-i} \times 5)C$ to $2^{3-i}C$.

IV. Experiments

A. Datasets

In our experiments, we used four public CD datasets: CDD dataset, WHU-CD dataset, OSCD dataset, and HRSCD dataset.

The CDD [44] dataset contains 11 pairs of bitemporal images with seasonal changes, which include 7 pairs of images with the sizes of 4725×2200 pixels and 4 pairs of images with the sizes of 1900×1000 pixels. These images were taken by Google Earth, and their spatial resolutions are from 3 to 100 cm/pixel. In the CDD dataset, the changes that need to be detected include the changes caused by buildings, cars, roads, and so on, but not include the changes caused by seasons. In [43], all images were partitioned into 256×256 image patches by cropping and rotating, and 16 000 pairs of patches were generated. For these patches, 10 000 pairs are used for training, 3000 pairs are used for validation, and the remaining are used for testing.

The WHU-CD [45] dataset includes two aerial images with 0.075-m spatial resolution, both of which were taken in Christchurch, but at different times, one in 2012 and the other in 2016. In 2011, there was a magnitude 6.3 earthquake in Christchurch, and the WHU-CD dataset recorded the reconstruction situation of this city after that earthquake. In the WHU-CD dataset, the changes that need to detect only include the changes caused by buildings. For the original images whose sizes are $32\,507 \times 15\,354$ pixels, we crop them into 224×224 nonoverlapping image patches. Following the previous method [22], we randomly divided these image patches into training set, verification set, and test set in an 8:1:1 ratio.

The OSCD [46] dataset includes 24 pairs of images collected by Sentinel-2. Each image has 13 bands, and only RGB three bands are used in our experiment. The size of these images is about 600×600, and the spatial resolution is 10 m. The changes annotated in this dataset are mainly urban changes, such as urban growth. Consistent with [46], we used 14 pairs of images for training and the rest for testing.

The HRSCD [47] dataset contains 291 pairs of $10\,000 \times 10\,000$ aerial images with a spatial resolution of 50 cm/pixel. The first image in each pair was taken in 2005 or 2006 and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

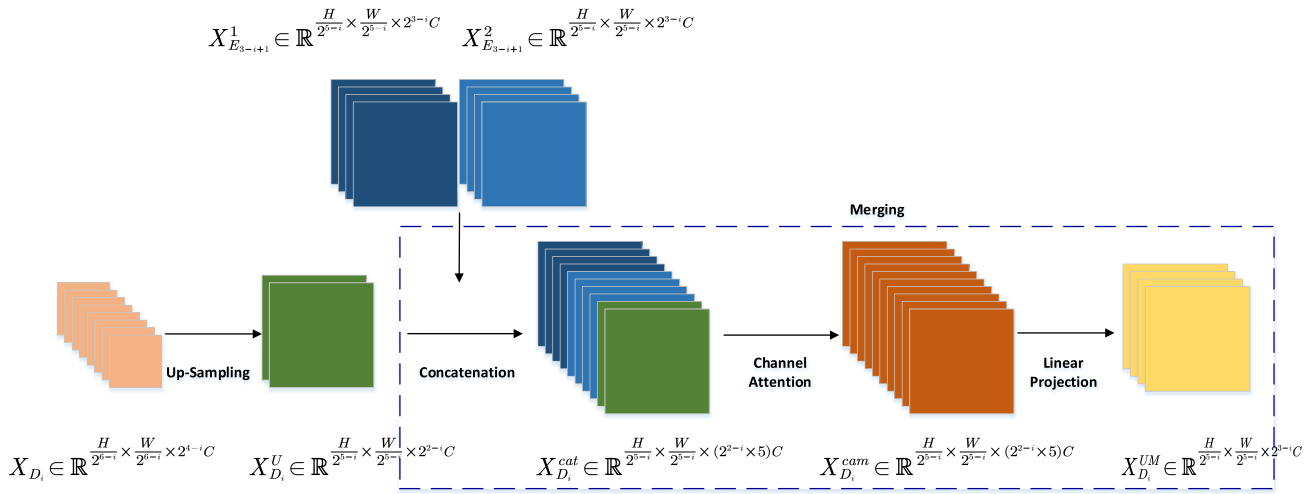


Fig. 5. UM block.

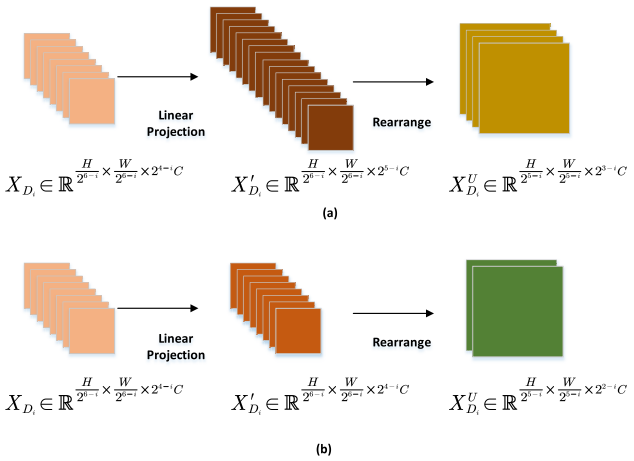


Fig. 6. Upsampling methods. (a) Patch expanding. (b) Patch reshaping.

the second in 2012. In our experiments, 126 pairs were used for training, 20 pairs for verification, and the rest for testing. This dataset is not only labeled with change information but also with terrain information.

B. Evaluation Criteria

We use eight evaluation indicators to evaluate CD results, which are TP, TN, FP, FN, precision, recall, $F1$-score, and overall accuracy (OA). In these evaluation indicators, $F1$-score is the most important one.

TP, TN, FP, and FN represent the number of true positive, true negative, false positive, and false negative, respectively. The calculation formulas of precision, recall, $F1$-score, and OA are given as follows:

$$\begin{aligned} \text{precision} &= \text{TP}/(\text{TP}+\text{FP}) \\ \text{recall} &= \text{TP}/(\text{TP}+\text{FN}) \\ F1 &= \frac{2}{\text{precision}^{-1}+\text{recall}^{-1}} \\ \text{OA} &= (\text{TP}+\text{TN})/(\text{TP}+\text{TN}+\text{FN}+\text{FP}). \end{aligned} \tag{3}$$

C. Implementation Details

In our experiments, we used a Swin transformer model pretrained on ImageNet to initialize the parameters of SwinSUNet. This pretrained model is provided in [32]. Then, we fine-tuned the model on the CD dataset to further optimize the parameters.

In the training of SwinSUNet, the deep learning framework we used was PyTorch, and the graphics card type we used was Nvidia Titan XP. Some important training settings of our experiments are described as follows: the network optimizer was the Adam algorithm, the epoch number was 200, and the learning rate was 0.00001. For data augmentation, we only performed a simple flip operation on the training data. At each epoch, we calculated the model's $F1$-score on the verification set. At the end of training, we got the best model, which had the highest $F1$-score. Then, we used this model to evaluate the test set.

D. Comparative Experiments

1) Comparative Methods: We compared SwinSUNet with several excellent methods, such as CDNet [48], FC-EF [20], FC-Siam-conc [20], FC-Siam-diff [20], DASNet [22], and STANet [21].

CDNet [48] is a simple U-shaped CD network, which contains one contraction module and one expansion module. It needs to concatenate two three-channel bitemporal images into a six-channel matrix and then uses the concatenated matrix as the input.

Like CDNet [48], FC-EF [20] has a U-shaped structure and uses concatenated images as the inputs. However, FC-EF adds some skip connection layers to the network.

FC-Siam-conc [20] has a Siamese U-shaped structure. The encoder of FC-Siam-conc can process bitemporal images in parallel and generate bitemporal features at various levels. For the bitemporal features at the same level, the encoder further uses the concatenation operation to fuse them. The decoder of FC-Siam-conc leverages a hierarchy structure to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON RESULTS ON THE CDD TEST SET

Method	Precision	Recall	F1	OA	TP	TN	FP	FN
FC-EF[20]	0.905	0.421	0.574	0.919	10690737	170079741	1117020	14720502
FC-Siam-conc [20]	0.918	0.505	0.652	0.930	12836061	170048201	1148560	12575178
FC-Siam-diff [20]	0.915	0.408	0.564	0.919	10365871	**170230994**	**965767**	15045368
CDNet[48]	0.930	0.799	0.860	0.966	20307786	169663217	1533544	5103453
DASNet[22]	0.920	0.930	0.925	0.981	23643151	169145108	2051653	1768088
STANet[21]	0.931	**0.939**	0.935	0.983	**23851762**	169428064	1768697	**1559477**
SwinSUNet	**0.957**	0.923	**0.940**	**0.985**	23455616	170150627	1046134	1955623

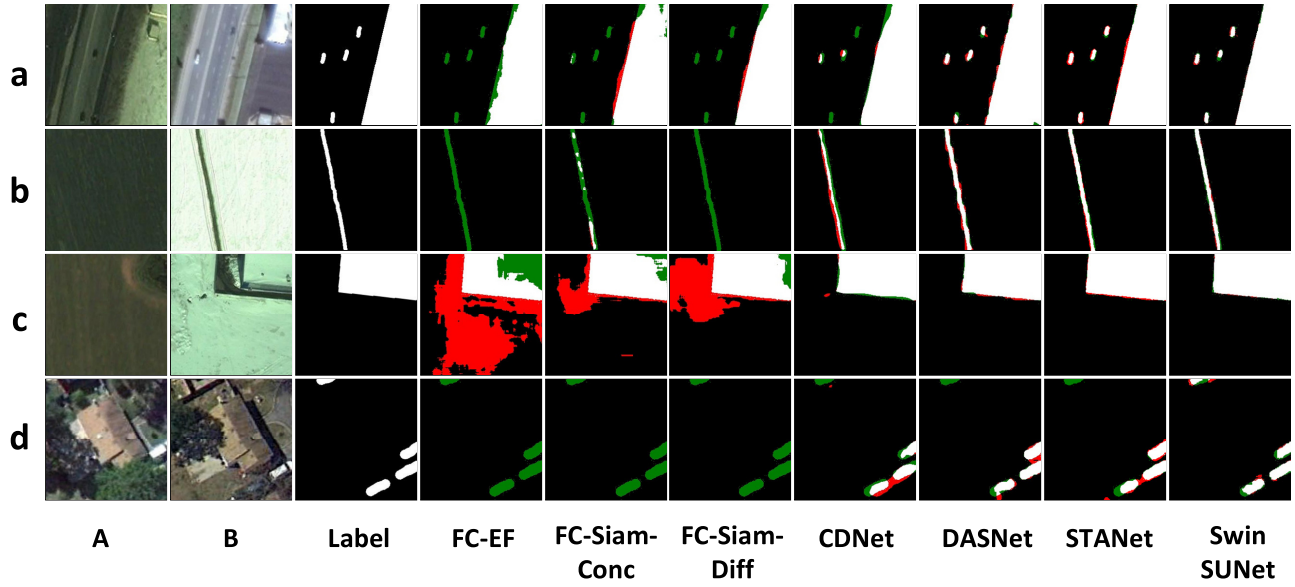


Fig. 7. Visualization results of several CD methods on CDD test set. For the convenience of viewing, we use different colors to express different meanings; white means true positive, black means true negative, red means false positive, and green means false negative. (a)–(d) Four sets of images randomly selected from the CDD test set, respectively.

gradually generate change information and eventually restores the change maps.

Like FC-Siam-conc [20], FC-Siam-diff [20] has a Siamese U-shaped structure. The difference is that FC-Siam-diff uses absolute difference operation to merge the bitemporal features.

DASNet [22] first extracts semantic features by using Resnet50. Second, it further processes the features by using a dual attention module. Finally, it generates the change maps by using a metric module. The dual attention module can extract features that contain both channel information and spatial information.

STANet [21] uses Resnet18 to extract features and then leverages a pyramid spatial–temporal attention module to further process the features. Finally, it uses a metric module to generate the change maps. The pyramid spatial–temporal attention module is used to cope with the changes at different scales.

In our experiments, in addition to CDNet, we reimplemented the above algorithms by using the source codes provided by the authors. For CDNet, we reimplemented it according to the description in this article.

2) Experimental Results on CDD Dataset: Table I presents the results of several methods on the CDD dataset. From this table, we can conclude that among all the experimental methods, SwinSUNet achieves the highest score on three evaluation indicators. In particular, the $F1$-score of our SwinSUNet is 94.0%, which outperforms STANet by 0.5%.

We selected several images from the CDD test set and used various algorithms to generate the change maps. Fig. 7 shows the generated change maps. From these visualization results, it is easy to see that the change maps generated by SwinSUNet have better details and are closer to the ground truth than other methods.

Fig. 7(a) contains the changes caused by cars and buildings, Fig. 7(b) contains the changes caused by roads, and Fig. 7(c) only contains the changes caused by buildings. From these figures, we can see that SwinSUNet produced fewer detection errors and predicted more accurate edges than other methods. Fig. 7(d) contains the changes caused by cars, and there exists occlusion in it. From the figures, we can see that many methods failed to detect the changes caused by the cars under the leaves, but SwinSUNet was still able to detect these changes accurately.

3) Experimental Results on WHU-CD Dataset: Table II presents the CD results of various methods on the WHU-CD

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON RESULTS ON WHU-CD TEST SET

Method	Precision	Recall	F1	OA	TP	TN	FP	FN
FC-EF[20]	0.706	0.722	0.714	0.972	1737816	46394060	723211	668625
FC-Siam-conc[20]	0.474	0.712	0.569	0.948	1712205	45213977	1903294	694236
FC-Siam-diff[20]	0.763	0.666	0.712	0.974	1603014	46620718	496553	803427
CDNet[48]	0.822	0.856	0.838	0.984	2059558	46670607	446664	346883
DASNet[22]	0.905	**0.926**	0.916	0.992	2228406	46884084	233187	178035
STANet[21]	0.944	0.873	0.907	0.991	2100299	46992924	124347	306142
SwinSUNet	**0.950**	**0.926**	**0.938**	**0.994**	**2228689**	**47000829**	**116442**	**177752**

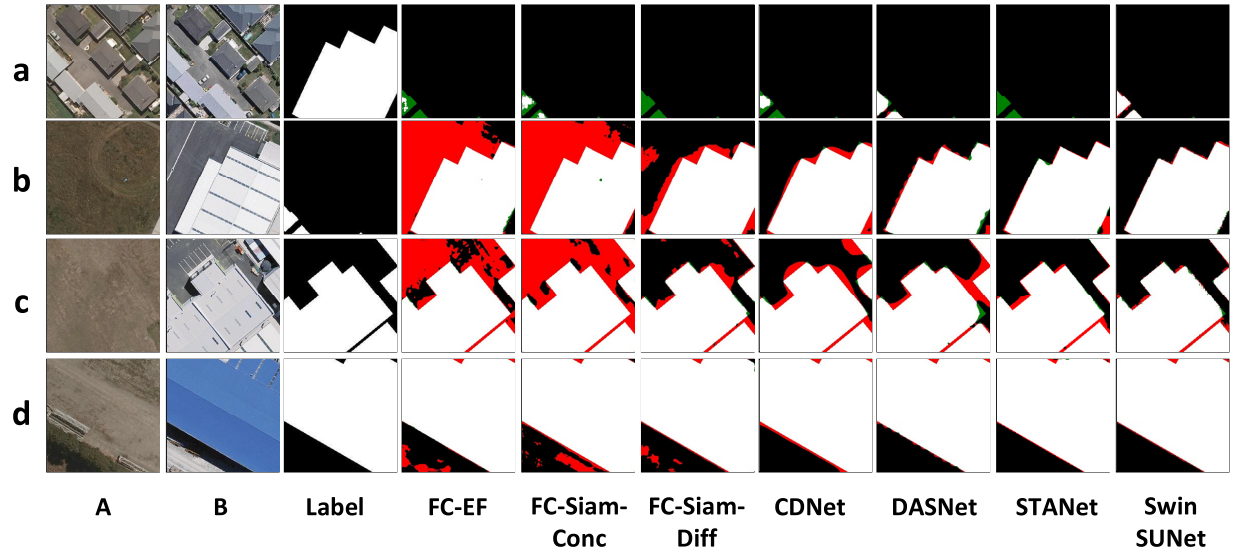


Fig. 8. Visualization results of several CD methods on the WHU-CD test set. The meanings of the different colors are the same as in Fig. 7. (a)–(d) Four sets of images randomly selected from the WHU-CD test set, respectively.

TABLE III
COMPARISON RESULTS ON THE OSCD TEST SET

Method	Precision	Recall	F1	OA	TP	TN	FP	FN
FC-EF[20]	**0.692**	0.436	0.535	**0.961**	69434	**2888004**	**30855**	89643
FC-Siam-conc[20]	0.387	0.549	0.454	0.932	87313	2780819	138040	71764
FC-Siam-diff[20]	0.437	0.396	0.415	0.942	62969	2837638	81221	96108
CDNet[48]	0.482	**0.561**	0.519	0.946	**89312**	2822926	95933	**69765**
SwinSUNet	0.550	0.540	**0.545**	0.953	85936	2848553	70306	73141

dataset. From this table, we can conclude that among all the experimental methods, SwinSUNet achieves the optimal results on all evaluation indicators. In particular, the $F1$-score of our SwinSUNet is 93.8% and outperforms DASNet by 2.2%.

For some images that are selected from the WHU-CD test set, various algorithms are used to generate the change maps. The visualization results are shown in Fig. 8. From these figures, we can see that the edge predicted by SwinSUNet is more accurate than other methods.

The changes in Fig. 8 are all caused by buildings. For a small change in Fig. 8(a), many algorithms failed to detect it, but SwinSUNet could still detect it. From the visualization results in Fig. 8(b)–(d), it is easy to see that the detection results of SwinSUNet are closer to the ground truth and the edges predicted by SwinSUNet are more accurate than other methods.

4) Experimental Results on OSCD Dataset: The images in the OSCD dataset have 13 bands. In the experiments, we only used RGB three bands. Compared to other datasets, the OSCD dataset has a very small number of images, so it is very difficult to obtain good CD results. The experimental results of various methods are shown in Table III. We can see that SwinSUNet achieved the highest $F1$-score of all the methods. Fig. 9 shows the visualization results of multiple methods on the OSCD dataset.

5) Experimental Results on HRSCD Dataset: In the HRSCD dataset, the proportion of unchanged pixels is very high, 99.232%. Thus, this dataset is very challenging. The results of various algorithms on the HRSCD dataset are shown

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

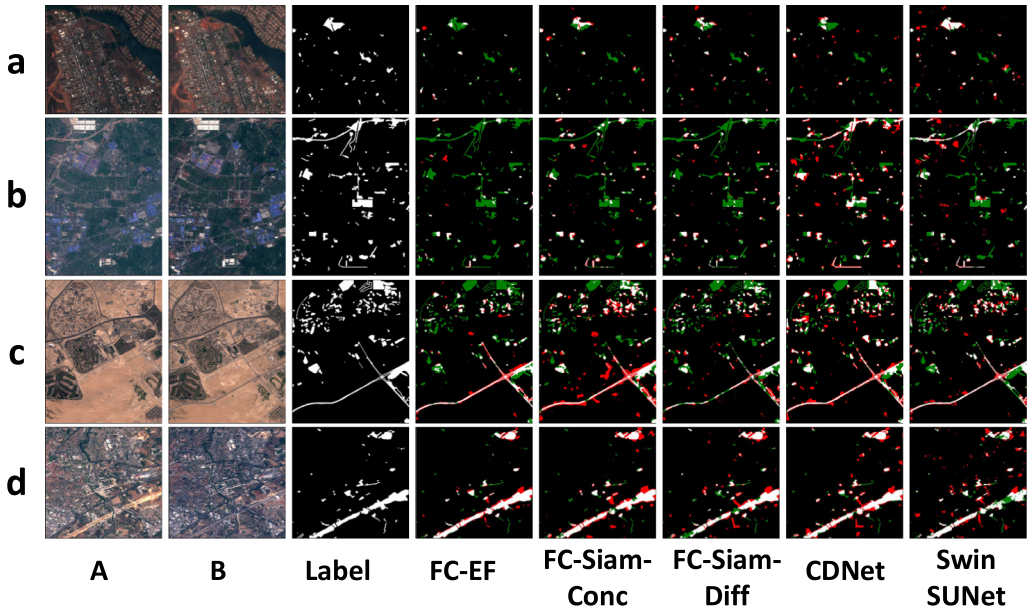


Fig. 9. Visualization results of several CD methods on the OSCD test set. The meanings of the different colors are the same as in Fig. 7. (a)–(d) Four sets of images randomly selected from the OSCD test set, respectively.

TABLE IV
COMPARISON RESULTS ON THE HRSCD TEST SET

Method	Precision	Recall	F1	OA	TP	TN	FP	FN
FC-EF[20]	0.228	0.124	0.161	**0.993**	9910119	**13874881265**	**33637113**	69837487
FC-Siam-conc[20]	0.175	0.120	0.142	0.992	9567276	13863300665	45217713	70180330
FC-Siam-diff[20]	0.151	0.240	0.185	0.988	19134813	13800904535	107613843	60612793
CDNet[48]	0.229	0.321	0.267	0.990	25591904	13822231999	86286379	54155702
DASNet[22]	0.229	0.334	0.272	0.990	26675304	13818743833	89774545	53072302
STANet[21]	**0.258**	0.218	0.236	0.992	17356815	13858719813	49798565	62390791
SwinSUNet	0.243	**0.343**	**0.285**	0.990	**27354150**	13823459477	85058901	**52393456**

in Table IV. As it is possible to see in the table, the OA value of many algorithms is very high, but $F1$-score is very low, such as FC-EF, FC-Siam-conc, and STANet. Therefore, it is very difficult to obtain high OA and $F1$-score at the same time. SwinSUNet achieved the highest $F1$-score and also achieved a good OA value. SwinSUNet is proven to be effective. It should be noted that, in keeping with our previous experiments, we did not use the terrain label but only the change label.

E. Parameter Analysis

In the process of training, parameters usually have a great influence on the final model. Thus, choosing the appropriate training parameters is very important. In this section, we mainly discuss the influence of epoch number and learning rate on SwinSUNet.

1) Epoch Number: Fig. 10 shows the line chart generated during SwinSUNet training. Fig. 10(a) shows the test results on the CDD validation set, and Fig. 10(b) shows the test results on the WHU-CD validation set. As can be seen from these figures, the model will converge when it reaches a certain epoch. Too much epoch will not significantly increase the CD capability of the model. Therefore, in our experiments, the number of epochs we selected was 200.

2) Learning Rate: In order to obtain an appropriate learning rate, we trained three SwinSUNet models on the CDD dataset. The three models used different learning rates during training, which were 0.0001, 0.00001, and 0.000001. The experimental results are shown in Table V. Obviously, the model with a learning rate of 0.00001 obtained the best results.

F. Ablation Study

It is well known that many factors can affect the model results, such as network structure and parameter initialization method. In this section, we mainly research the influences of pretraining, UM block, and model depth on the SwinSUNet model. For these three factors, we conducted ablation experiments on the CDD dataset.

1) Impact of UM Block: The UM block contains a patch reshaping block and a merging block. In experiments, we verify the validity of these two blocks.

Cao *et al.* [42] designed patch expanding as the upsampling method. We make a little change to patch expanding and name the new upsampling method as patch reshaping. The only difference between these two upsampling methods is that the linear projection layer in patch reshaping does not increase

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

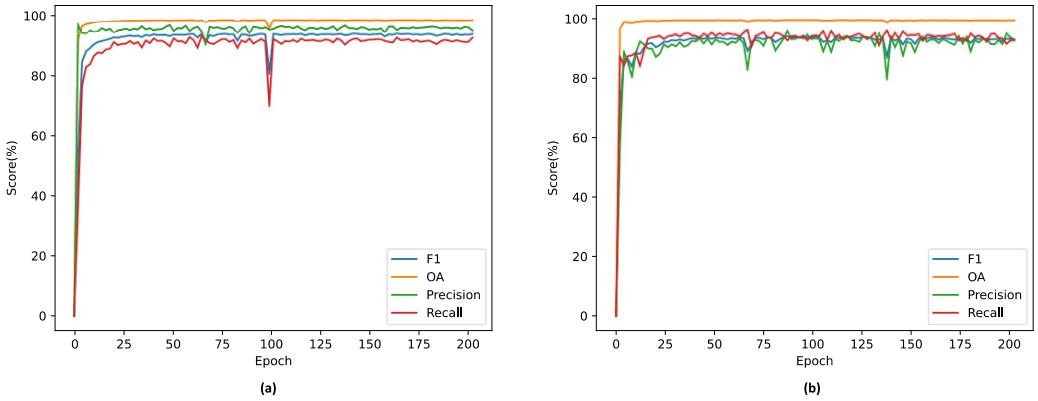


Fig. 10. Effect of epoch number on (a) CDD validation set and (b) WHU-CD validation set.

TABLE V
LEARNING RATE ANALYSIS

Learning rate	Precision	Recall	F1	OA	TP	TN	FP	FN
0.0001	0.932	0.897	0.914	0.978	22797428	169535821	1660940	2613811
0.00001	**0.957**	**0.923**	**0.940**	**0.985**	**23455616**	**170150627**	**1046134**	**1955623**
0.000001	0.954	0.917	0.935	0.984	23307882	170070684	1126077	2103357

TABLE VI
U|M BLOCK IMPACT

Method	Precision	Recall	F1	OA	TP	TN	FP	FN
Transposed convolution	0.954	0.920	0.937	0.984	23382126	170056698	1140063	2029113
Patch expand	0.951	0.920	0.935	0.983	23380793	169982401	1214360	2030446
Patch reshape	0.952	**0.927**	0.939	0.984	**23550825**	170007308	1189453	**1860414**
Patch reshape + CAM	**0.957**	0.923	**0.940**	**0.985**	23455616	**170150627**	**1046134**	1955623

TABLE VII
MODEL DEPTH ANALYSIS

Configuration	Precision	Recall	F1	OA	TP	TN	FP	FN
Conf.1	**0.960**	0.905	0.932	0.983	23003290	**170236570**	**960191**	2407949
Conf.2	0.946	**0.931**	0.938	0.984	**23658505**	169846780	1349981	**1752734**
Conf.3	0.957	0.923	**0.940**	**0.985**	23455616	170150627	1046134	1955623

TABLE VIII
IMPACT OF PRETRAINING

Pre-trained	Precision	Recall	F1	OA	TP	TN	FP	FN
No pre-trained	0.939	0.841	0.888	0.972	21377756	169820596	1376165	4033483
Pre-trained	**0.957**	**0.923**	**0.940**	**0.985**	**23455616**	**170150627**	**1046134**	**1955623**

the dimension of the feature map. In this article, for verifying the effectiveness of patch reshaping, we compare it with other two upsampling methods: transposed convolution and patch expanding. For these three upsampling methods, we conducted experiments on the CDD dataset. The experimental results are shown in Table VI. The experimental results show that SwinSUNet with patch reshaping achieves better CD performance than the other two.

Different from the previous methods, we add a CAM in the merging block. The purpose of adding the CAM is to integrate features more effectively in the channel dimension. We proved the effectiveness of adding CAM through experiments. The results of the experiment are shown in Table VI.

2) Analysis of Model Depth: As everyone knows, the model depth seriously affects the model performance. In our experiments, three model depth configurations were used. Conf.1 is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

(2 2 2 2 2 2), Conf.2 is (2 2 6 2 2 2 2), and Conf.3 is (2 2 6 2 6 2 2). The meanings of the values in the configurations are given as follows. The first three values represent the number of Swin transformer blocks in each stage of encoder, the fourth value represents the number of Swin transformer blocks in fusion, and the last three values represent the number of Swin transformer blocks in each stage of decoder.

In the experiments, we test these three configurations on the CDD dataset. The experimental results are shown in Table VII. It is clear that as the number of Swin transformer blocks increases, the performance of CD improves. However, the more Swin transformer blocks there are, the more computations the model requires. In this article, the model depth configuration used is Conf.3.

3) Impact of Pretraining: Before training, we used a Swin transformer model pretrained on the ImageNet dataset to initialize the parameters of SwinSUNet. For comparison and verifying the necessity of pretraining, we also trained a SwinSUNet model from scratch. We conducted experiments on the CDD dataset, and the results of these two training ways are shown in Table VIII. From these results, it is easy to see that the score of the pretraining is higher than no pretraining in all evaluating indicators. In particular, the $F1$-score of the former is 5.2% higher than that of the latter. Therefore, pretraining is very important for the training of SwinSUNet.

V. Conclusion

In order to explore the potential of pure transformer network in the field of CD, this article proposes SwinSUNet, a pure Swin transformer network with Siamese U-shaped structure. This is the first CD network based on pure transformer. Different from the previous CNN-based CD methods, SwinSUNet uses the Swin transformer block as the basic unit. The transformer can extract long-term global information in space-time, so SwinSUNet can better extract change information than other CNN-based methods.

In our experiments, we used four public datasets: CDD dataset, WHU-CD dataset, OSCD dataset, and HRSCD dataset. To verify the effectiveness of SwinSUNet, we did expensive experiments on these datasets. From the experimental results, it is directly to see that SwinSUNet is significantly better than the other methods. Through experiments, the effectiveness of SwinSUNet is proven; meanwhile, the huge potential of pure transformer network for the CD task is confirmed.

References

[1] X. Huang, L. Zhang, and T. Zhu, "Building change detection from multitemporal high-resolution remotely sensed images based on a morphological building index," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 105–115, Jan. 2014.

[2] C. Marin, F. Bovolo, and L. Bruzzone, "Building change detection in multitemporal very high resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2664–2682, May 2015.

[3] P. Coppin, I. Jonckheere, K. Nackaerts, B. Muys, and E. Lambin, "Digital change detection methods in ecosystem monitoring: A review," *Int. J. Remote Sens.*, vol. 25, no. 9, pp. 1565–1596, 2004.

[4] K. Rokni, A. Ahmad, A. Selamat, and S. Hazini, "Water feature extraction and change detection using multitemporal landsat imagery," *Remote Sens.*, vol. 6, no. 5, pp. 4173–4189, May 2014.

[5] J. Xu, W. Lu, Z. Li, P. Khaitan, and V. Zaytseva, "Building damage detection in satellite imagery using convolutional neural networks," 2019, *arXiv:1910.06444*.

[6] C.-F. Chen *et al.*, "Multi-decadal mangrove forest change detection and prediction in Honduras, Central America, with landsat imagery and a Markov chain model," *Remote Sens.*, vol. 5, no. 12, pp. 6408–6426, Nov. 2013.

[7] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[8] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[9] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

[10] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[11] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask r-cnn," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2980–2988.

[12] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[13] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[14] H. Noh, S. Hong, and B. Han, "Learning deconvolution network for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1520–1528.

[15] Y. Sun, X. Wang, and X. Tang, "Deep learning face representation from predicting 10,000 classes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1891–1898.

[16] Y. Wen *et al.*, "A discriminative feature learning approach for deep face recognition," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 499–515.

[17] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4690–4699.

[18] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.

[19] A. Fujita, K. Sakurada, T. Imaizumi, R. Ito, S. Hikosaka, and R. Nakamura, "Damage detection from aerial images via convolutional neural networks," in *Proc. 15th IAPR Int. Conf. Mach. Vis. Appl. (MVA)*, May 2017, pp. 5–8.

[20] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[21] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[22] J. Chen *et al.*, "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[23] M. Zhang, G. Xu, K. Chen, M. Yan, and X. Sun, "Triplet-based semantic relation learning for aerial remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 266–270, Feb. 2019.

[24] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[25] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[26] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Sep. 2021.

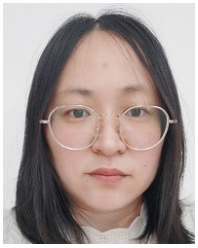
[27] H. Chen, Z. Qi, and Z. Shi, "Efficient transformer based method for remote sensing image change detection," 2021, *arXiv:2103.00208*.

[28] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.: Annu. Conf. Neural Inf. Process. Syst.*, Dec. 2017, pp. 5998–6008.

[29] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.

[30] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Mach. Learn.* PMLR, 2021, pp. 10347–10357.
[31] B. Wu *et al.*, "Visual transformers: Token-based image representation and processing for computer vision," 2020, *arXiv:2006.03677*.
[32] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted Windows," 2021, *arXiv:2103.14030*.
[33] D. Zhang, H. Zhang, J. Tang, M. Wang, X. Hua, and Q. Sun, "Feature pyramid transformer," in *Computer Vision—ECCV 2020*, vol. 12373, A. Vedaldi, H. Bischof, T. Brox, and J.-M. Frahm, Eds. Glasgow, U.K.: Springer, 2020, pp. 323–339.
[34] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Computer Vision—ECCV 2020*, vol. 12346, A. Vedaldi, H. Bischof, T. Brox, and J.-M. Frahm, Eds. Glasgow, U.K.: Springer, 2020, pp. 213–229.
[35] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.
[36] S. Zheng *et al.*, "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.
[37] M. Chen *et al.*, "Generative pretraining from pixels," in *Proc. 37th Int. Conf. Mach. Learn. (ICML)*, vol. 119, Jul. 2020, pp. 1691–1703.
[38] P. Esser, R. Rombach, and B. Ommer, "Taming transformers for high-resolution image synthesis," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12873–12883.
[39] F. Yang, H. Yang, J. Fu, H. Lu, and B. Guo, "Learning texture transformer network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 5791–5800.
[40] H. Chen *et al.*, "Pre-trained image processing transformer," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12299–12310.
[41] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 815–823.
[42] H. Cao, Y. Wang, J. Chen, D. Jiang, and M. Wang, "Swin-Unet: Unet-like pure transformer for medical image segmentation," 2021, *arXiv:2105.05537*.
[43] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Apr. 2020.
[44] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vols. 42, pp. 565–571, May 2018.
[45] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[46] R. C. Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Urban change detection for multispectral Earth observation using convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 2115–2118.
[47] R. Caye Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Multitask learning for large-scale semantic change detection," *Comput. Vis. Image Understand.*, vol. 187, Oct. 2019, Art. no. 102783.
[48] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Auto. Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.



Cui Zhang received the master's degree from the Faculty of Information Science and Engineering, Ocean University of China, Qingdao, China, in 2011. She is currently pursuing the Ph.D. degree with the School of Information Science and Engineering, Xinjiang University, Ürümqi, China.

Her research interests include computer vision and remote sensing image change detection.



Liejun Wang received the Ph.D. degree from the School of Information and Communication Engineering, Xi'an Jiaotong University, Xi'an, China, in 2012.

He is currently a Professor with the School of Information Science and Engineering, Xinjiang University, Ürümqi, China. His research interests include wireless sensor networks, computer vision, and natural language processing.



Shuli Cheng received the Ph.D. degree from the School of Information and Engineering, Xinjiang University, Ürümqi, China, in 2021.

He is currently working with the School of Information Science and Engineering and a Post-Doctoral Researcher with the College of Mathematics and System Science, Xinjiang University. His research interests include computer vision, natural language processing, and multimodal processing.



Yongming Li received the master's degree from the School of Information and Engineering, Xinjiang University, Ürümqi, China, in 2007.

He is currently an Associate Professor with the School of Information Science and Engineering, Xinjiang University. His research interests include computer vision, natural language processing, and wireless sensor networks.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 01:23:20 UTC from IEEE Xplore. Restrictions apply.